MARC21 Bibliographic to LRM/RDA/RDF Mapping and Conversion Project

Phase I Project Summary

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Project Background

Rationale

The MARC21 Bibliographic to LRM/RDA/RDF Mapping and Conversion Project (MARC2RDA) began in response to a specific set of problems facing the library metadata community. Much of the world's bibliographic metadata is formatted as MARC21. MARC datasets are siloed based on storage databases and the languages used by its creators, among other factors. MARC21's reliance on text strings rather than machine-readable statements limits its usefulness in the Semantic Web. Libraries are currently using several linked data ontologies with conflicting data models to experiment with moving MARC21 data into a linked data implementation scenario. Many of these ontologies are not compatible with the International Federation of Library Associations and Institutions (IFLA) Library Reference Model (LRM), compounding challenges for libraries planning to implement Official RDA in the near future.

The IFLA LRM is an internationally recognized conceptual model for library bibliographic metadata. RDA (Resource Description and Access) is an implementation of the IFLA LRM, and the RDA Steering Committee (RSC) publishes an open source ontology via the RDA Registry which can be used to represent bibliographic metadata in alignment with both Official RDA and the IFLA LRM. For the sake of reducing silos and increasing interoperability, the use of the LRM/RDA/RDF ontology in the representation of international bibliographic metadata is desirable.

Goals

To facilitate tool development for a linked data implementation of RDA, the Project set out to achieve the following goals:

Primary Goals

- Create an open source, robust mapping between MARC21 Bibliographic and LRM/RDA/RDF standards
- Facilitate the move from record-based cataloging (text strings) to entity-based cataloging (semantic things) by identifying, relating, and describing appropriate RDA entities using legacy MARC21 Bibliographic records
- Create a corresponding open source transformation tool (the The MARC2RDA Project Transform) using XSLT, including:
    - A downloadable transformation code package for converting large files
    - A web-based application for converting small files (up to 20MB)

- Openly publish a large sample dataset of transformed legacy MARC21 Bibliographic records, including:
  - A subset of the Library of Congress catalog
  - A subset of the National Library of New Zealand catalog
  - Original records from the University of Washington catalog

Secondary Goals

- Provide feedback directly to the RSC on the documentation available through the Official RDA Toolkit and RDA Registry
- Provide feedback directly to the Program for Cooperative Cataloging (PCC) on its RDA and linked data policies and practices
- Maintain project tools as the MARC21 and LRM/RDA/RDF standards evolve
- Maintain clear documentation to support the implementation of the LRM/RDA/RDF ontology

Strategic Approach

As a record-based and string-heavy format, MARC21 does not explicitly define the bibliographic entities it describes. In order to transform MARC21 data into entity-based descriptions suitable for a linked data implementation scenario, it is necessary to identify, describe, and relate all of the entities which are implied to exist in MARC21 records as exemplars of RDF classes. In terms of the LRM and RDA, MARC21 records carry data about entities which fall into two categories: resource entities and related entities.

Resource entities described by MARC21 records include Works, Expressions, Manifestations, and Items (WEMI). Related entities described by MARC21 records include Agents, Persons, Corporate Bodies, Families, Works, Expressions, Manifestations, Items, Nomens, Places, and Timespans. To identify and describe resource and related entities, information must be extracted from coded, descriptive, and headings fields in MARC21 records. To relate entities to one another, relationship data must be gleaned primarily from headings fields in MARC21.

Descriptive data and relationships which can't be reliably mapped to a specific RDA entity have been mapped as Manifestation elements and relationships in most cases because MARC21 records are generally created at the Manifestation level.

Project Phases and Deliverables

The MARC21 Bibliographic Standard features hundreds of fields and thousands of subfields and fixed field positions, and is built to accommodate all resource formats which might find their way into a GLAM (gallery, library, archive, or museum) catalog. The Project divided its work into manageable pieces in order to complete its work iteratively, in phases which are practically useful on their own but are meant to build on one another.

Phase I Scope, Timeline, and Deliverables

Scope

Phase I of MARC2RDA maps all fields considered core in the BIBCO Standard Record (BSR), and features XSLT transformation code for all of these fields. A large transformation including the datasets of several national libraries is run, with resulting data openly published in a GraphDB triple store. All output data, mapping documentation, and transformation code is published openly.

Aggregates and reproductions significantly complicate the identification, description, and relation of bibliographic entities from MARC21 legacy data. To allow for more time to address complexities, Phase I features rudimentary treatment of some aggregates and reproductions meant to be built upon in future phases. Phase I includes logical patterns and XSLT coding to distinguish Group 1 (simple augmentation aggregate manifestations and non-aggregate manifestations) from Group 2 (complex augmentation, parallel, and collection aggregate manifestations, and diachronic and collection works). Group 1 resources are transformed as part of Phase I, and Group 2 resources are excluded and left for Phase II.

Timeline

MARC2RDA Phase I began March 3, 2021 and ended September 30, 2025. Initial mapping of the BSR was completed February 28, 2025. Mapping review was finalized March 31, 2025. The transformation code was completed August 30, 2025 and finalized for Phase I on September 30, 2025. Deliverables were made public September 30, 2025. Due to changes in volunteer time commitments and project scope, the timeline for Phase I of the Project has been adjusted several times.

Community feedback and subsequent revisions to mappings and code are anticipated, and will be addressed during future phases.

Deliverables

Phase I deliverables are described in detail below. See: Phase I Release.

Phase II Scope, Timeline, and Deliverables

Scope

Phase II will include all fields considered core in the BSR and CONSER Standard Record (CSR). Fields will be robustly mapped and transformed, with mappings, transformation code, and output published openly. Entity deduplication will be addressed more seriously in this phase, and Group II resources will be included. The possibility of lookups to external data sources such as FAST will be explored. Test output data will be re-published in the GraphDB triple store. Guidelines will be developed to encourage pre- and post-processing of data. The project will communicate with relevant organizations and stakeholders to encourage their participation in developing and maintaining project deliverables.

Timeline

The current projected timeline for Phase II is approximately two years, beginning November 1, 2025.

Deliverables

The Project will publish deliverables as part of an official GitHub release of its repository. Revised mapping spreadsheets, transformation code, and output in GraphDB will be provided. Documentation similar to that of Phase I will be included in the GitHub release.

Summary of Phase I Release

GitHub Release

An official GitHub release featuring README files at multiple levels to guide use and interpretation was published at the end of September 2025. The release includes transformation code in the form of XSLT stylesheets along with accompanying documentation and brief guidelines for pre- and post-processing output data. Phase I mappings are published as .csv files. Documentation in the form of written summaries is included, as well as an index facilitating easy navigation of deliverables.

Materials accompanying the GitHub release include sample output data in the form of a GraphDB triple store, additional vocabularies published by the University of Washington Libraries, a list of Approved URIs, and feedback directly submitted to the RSC and PCC.

Major Benchmarks

Entity Identification, Description, and Relation

Identifying Resource Entities

Described Manifestation

Every MARC21 Bibliographic record describes a manifestation as a resource entity. *There are exceptions to this, e.g., records that are used to describe both an original record and a reproduction*.

Each described manifestation embodies at least one expression and one work.

The Project has developed a set of logical markers in order to identify different categories of resource descriptions that need to be processed using special coding. These are descriptions of:

- Collection works (the description is of a manifestation, expression, and work that is a collection of items)

- Diachronic works (the description is of a manifestation, expression, and work that has diachronic characteristics)

- Aggregate manifestations (the description is of a manifestation and one or more expressions and works that have aggregating characteristics)

- Single expression manifestations (the description is of a manifestation and only one expression and work)

Described expression

Every MARC21 Bibliographic record describes one or more than one expression as a resource entity.

Described work

Every MARC21 Bibliographic record describes one or more than one work as a resource entity.

- If aggregate markers are present, then the transformation code determines how many works and expressions are described

- If diachronic markers are present, the transformation code determines whether the work or works described by a record are static or diachronic.

- If collection work markers are present, the transformation code determines that the record describes a collection work

Described item

Some MARC21 Bibliographic records also provide item-specific information, sometimes in relationship subfields, often using subfield $5.

See treatment of subfield 5 for further information about item data related to items held in particular collections.

Access Points for Resource Entities

Access points are not strictly required by RDA, but serve useful purposes in cataloging and discovery workflows. While access points are occasionally explicitly provided in MARC21 for resource entities through fields such as 130 and added entry headings fields such as 710-730, they sometimes need to be devised by the transformation (e.g., combining fields 100+240).

IRIs for Resource Entities

If an IRI for a resource entity is provided in a subfield $1 or $0, then that IRI might be used for the entity (see the section on Minting IRI's, below).

If an IRI for a resource entity is not provided or cannot be used, then the transformation code creates an IRI for the entity using a set pattern based on its access point. This method for minting IRIs helps to deduplicate entities. This method of deduplication is error-prone, and is meant to be built upon in Phase II. See section below on "Minting IRIs" for more details.

Resource to Resource Entity Relationships

Resource entities have pre-determined relationships within each "WEMI stack" (set of primary works, expressions, manifestations, and items for a given resource). Inverse relationships are explicitly stated between resource entities.

Identifying Agent Entities

Resource entities are frequently related to agent entities (persons, families, corporate bodies) using headings fields in MARC21. The transformation code identifies such agents based on the MARC21 fields, indicators and subfields used for them.

Access Points for Agent Entities

The transformation code maps access points for Agents using the values provided in the appropriate MARC headings fields, indicators, and subfields. It does not attempt to construct access points for agents.

IRIs for Agent Entities

If agent entities are identified using IRIs from the Approved IRIs List in subfield $1 of a MARC21 field, those IRIs are reused to refer to agents in the transformation. If more statements need to be made about an agent or if subfield $1 does not contain a valid IRI for the agent, the code mints IRIs for agent entities in the same fashion as for resource entities (based on access points) in order to reduce duplication. See section below, "Minting IRIs", for more information. Further development is needed in this area to ensure that entity conflation and duplication are reduced.

Resource to Related Agent Entity Relationships

The transformation code uses an internally developed WEMI-Agent relationship table to identify relationship designators from appropriate MARC fields and subfields and maps the values found to corresponding RDA relationship elements. If any relationship values in the MARC records cannot be mapped, e.g., because they have no corresponding RDA relationship elements or because of typos, etc., the mapping defaults to related [Agent] to manifestation.

Identifying Concept Entities

MARC21 bibliographic records refer to many concepts which fall outside of the RDA Entity class data model. Such entities include classification numbers, topical subjects, genres, carrier types, modes of production, and more.

In MARC21 fields for concepts, the field and subfield coding or the contents of subfield $2 reliably identify sources and the relationships between concepts and resource entities.

For a document on how concept entities are modeled and transformed, see [Transforming subject data from MARC 21 to RDA]. Generally, concepts with IRIs are given as direct values of RDA elements as they are found. Concepts without valid IRIs are minted by the transformation as skos:Concepts and given sources and labels according to what is available in MARC21. See section below, "Minting IRIs", for more information.

Identifying Related WEMI Entities

Several fields in MARC21 Bibliographic records refer to works, expressions, manifestations, or items (WEMI) which are related to the main resource entities. Based on information encoded in MARC21, the Project aims to identify, describe, and relate as many of these entities as possible. Access points the project puts together for such entities are often less complete, as less information is available. IRIs are minted based on access points, and relationships are determined using the project's relator table and accompanying code. When creator agents are named for related WEMI entities, the project attempts to identify, describe, and relate them similarly to agents directly related to resource entities.

Resource to Related WEMI Entity Relationships

The transformation code uses an internally developed WEMI-WEMI relationship table to identify relationship designators from appropriate MARC fields and subfields and maps the values found to corresponding RDA relationship elements. If any relationship values in the MARC records cannot be mapped, e.g., because they have no corresponding RDA relationship elements or because of typos, etc., the mapping defaults to related [WEMI] to manifestation.

Retaining MARC data

For ease of output review and to preserve data provenance, the project includes the entire binary MARC input record in the output RDA/RDF, and relates it to the entities based on that record. This MARC is modeled as a note on a metadata manifestation, which manifests a metadata work. The metadata work is related to entities based on the record. Data provenance is an aspect of MARC21 bibliographic and RDA/RDF which will be built out in Phase II.

Timespan Entities and Date Strings

Periods of time related to resource entities and related entities are represented using structured and unstructured strings in MARC21 Bibliographic. Whether an RDA:Timespan entity is minted or not depends on whether there are additional statements to be made, such as a format/string encoding scheme, and also whether the date conforms to the definition of an RDA:Timespan (a finite period of time). Access points for Timespans are based on the strings used to represent them in MARC21, and IRIs are based on these access points in addition to their sources if available. Dates which are not made into Timespan entities are recorded as literal (string) values.

Where date strings or Timespans can be unambiguously related to an RDA entity, they are related to that entity in MARC2RDA according to the definitions of the MARC21 fields and subfields in which they appear. When date strings or Timespans could be related to either of two entities, they are generally related to the narrower entity (expression rather than work, for example), according to the best judgment of MARC2RDA participants.

Place Entities

Places related to resource entities can be identified in coded, headings, and descriptive fields in MARC21 data. They may be referred to using codes, structured and unstructured strings, or as linked data entities (in the form of IRIs). The MARC2RDA Project treats places as RDA:Place entities when something can be said about the entity beyond its label. When this is not the case, they are recorded directly as string values of appropriate elements as though they were appellations for an RDA:Place. Places in MARC21 are mapped to RDA elements on a field-by-field basis rather than through a table.

Nomen Entities

Nomens related to resource or related entities can be found in headings and descriptive fields in MARC21 data. They appear as structured strings or as linked data entities (in the form of IRIs). The MARC2RDA Project treats appellations for other RDA entities as RDA:Nomen entities when something can be said about the appellation beyond its string value. When this is not the case, they are recorded directly as string values of appropriate elements as though they were Nomen strings. Nomens in MARC21 are mapped to RDA elements on a field-by-field basis rather than through a table.

String Encoding Schemes

Structured values in RDA are literal (or string) values constructed according to a named set of rules or a syntax. When possible, Nomen strings are related to IRIs for their string encoding schemes using the element “scheme of nomen”.

Access Points

Access points and authorized access points may be recorded in RDA as either structured string values or as Nomen entities. Where a string encoding scheme may be pointed to, or where statements further than a label must be made, a Nomen entity is minted and its nomen string is related to an IRI for its string encoding scheme.

Where an access point has been authorized by an authority file, the MARC2RDA transformation code uses properties for authorized access points. If an access point is a variant, or if its use as an authorized access point in an authority file is unknown, the code uses properties for access points rather than authorized access points.

For the purpose of deduplication, MARC2RDA has endeavored to create unique access points for each entity produced by the transformation so far as it is possible.

Minting IRIs

Where entities in MARC21 Bibliographic are assigned IRIs from sources listed in our Approved URIs Table, those IRIs are used to refer to those entities. Where no IRI is present, or where the given IRI does not come from an approved source, the MARC2RDA transformation mints a new IRI for that entity.

Functions for minting IRIs are located at https://github.com/crystalyragui/MARC2RDA/blob/main/Working%20Documents/transformationCode/The MARC2RDA Project-iris.xsl. Along the IRI functions are functions that are called within these functions, such as lookups and processing string values, and identifier functions. IRIs that are minted during runtime are all generated with the same base, which can be input when the transform is run, with the BASE parameter. The default base IRI for testing is ‘http://marc2rda.info/testing/”, and for production purposes the base IRI is “http://marc2rda.info/transform/”.

Further characters in addition to the base IRI are added to entity IRIs according to the following logic. In these patterns, BASE = the base for the IRI, either the The Project’s base or one that is input as a custom IRI. AP = access point.

Main WEMI

Main single work or augmented work

{BASE}wor#{AP}

Example IRI for work: Glude, John B. NOAA aquaculture plan.
http://marc2rda.info/testing/wor#gludejohnbnoaaaquacultureplan

Main Expression

{BASE}exp#{AP}

Example IRI for expression: Glude, John B. NOAA aquaculture plan (Text : English)
http://marc2rda.info/testing/exp#gludejohnbnoaaaquacultureplantextenglish

Main Manifestation(s)

{BASE}{type}#{AP}

Type is "man" or "origMan" (manifestation, or original manifestation. This distinguishes reproduction manifestations from one another)

Example IRI for manifestation: NOAA aquaculture plan (1977 : United States Government Printing Office : Volume)
http://marc2rda.info/testing/man#noaaaquacultureplan1977unitedstatesgovernmentprintingofficevolume

Items:

{BASE}{RECORD}ite#{date at runtime}{generated_id}

For items, we mint a non-meaningful URI with a unique, opaque component.

Example: http://marc2rda.info/testing/ite#2025-02-05-0800d22e2106

All items are unique, we make no attempt to de-duplicate. We attempt to avoid duplicates on re-runs of the transformation by including the date at runtime.

Related RDA Entities

Related Agents & Works

Subfield $1 present from an approved source

If a subfield $1 is present and from an approved source, we use $1 as the IRI for the entity. In these cases, we do not describe the entity beyond an access point.

Example IRI from an approved subfield $1: https://orcid.org/0009-0008-8864-6143

Subfield $1 either absent or from unapproved source, and subfield $2 or indicators point to a source

If a subfield $2 is present AND a subfield $1 is either present but not from an approved source or not present, the transformation mints a new meaningful IRI for the entity.

{BASE}{$2_source}/{type}#{AP}

Example IRI with subfield $2 and unapproved or absent subfield $1:
*http://marc2rda.info/testing/*relwor*#gludejohnbnoaaaquacultureplan*

This approach allows for some automatic deduplication. Record identifiers are not included as part of the IRI–this value is only used when we want to mint IRIs which are deduplicated within but not between source records.

Types include “age” or “relWor”

For 6XX fields, sources may be in the second indicator rather than a $2. In those cases, those sources are used in IRIs.

Subfield $1 either absent or from unapproved source, and no source is indicated in $2 or indicators

If neither an approved subfield $1 or any $2 or source-identifying indicators is present, then we mint a meaningful IRI with no source provided

{BASE}{type}#{AP}

Related Places & Timespans

Subfield $1 present from an approved source

If a subfield $1 is present and from an approved source, we use $1 as the IRI for the entity. In these cases, we do not describe the entity beyond an access point.

Example IRI from an approved subfield $1: [*http://id.loc.gov/authorities/names/n79063919*](http://id.loc.gov/authorities/names/n79063919)

Subfield $1 either absent or from unapproved source, and subfield $2 or indicators point to a source

If a subfield $2 is present AND a subfield $1 is either present but not from an approved source or not present, the transformation mints a new meaningful IRI for the entity.

{BASE}{$2_source}/{type}#{AP}

Example IRI with subfield $2 and unapproved or absent subfield $1: [*http://marc2rda.info/testing/lcsh/pla#england*](http://marc2rda.info/testing/lcsh/pla#england)

This approach allows for some automatic deduplication.

Types include “pla” or “tim”

For 6XX fields, sources may be in the second indicator rather than a $2. In those cases, those sources are used in IRIs.

Subfield $1 either absent or from unapproved source, and no source is indicated in $2 or indicators

If neither an approved subfield $1 or any $2 or source-identifying indicators is present, then we mint a non-meaningful IRI with a unique, opaque component

{BASE}{type}#{date at runtime}{generated_id}

Related Nomens

When access points are from approved sources for the entity they are appellations for

We mint a meaningful IRI when the access point is approved for the entity named by the Nomen

{BASE}{source}/{type}#{AP}

Types may be "nom", "ageNom", "worNom", "plaNom", "timNom", etc.

Various 'types' of nomens are used to ensure unique nomen IRIs

Example IRI for a nomen from an approved source: http://marc2rda.info/testing/lcsh/plaNom#england

When access points are not from approved sources for the entity they are appellations for

Otherwise, we mint a non-meaningful IRI with a unique, opaque component

{BASE}{RECORD}{type}#{manifestation AP}{generated_id}

Example IRI for a nomen not from an approved source:
http://marc2rda.info/testing/worNom#2025-02-05-0800d22e17289

Types may be "nom", "ageNom", "worNom", "plaNom", "timNom", etc.

Various 'types' of nomens are used to ensure unique nomen IRIs

Related skos:Concepts

If a concept or subject does not have a source, an IRI is not minted and a string value is used. So, all skos:Concepts minted by the MARC2RDA transformation must have a source.

If a subfield $1 is present, this is used as a direct value.

If a subfield $0 is present and contains "http", this is used as a direct value.

If there is no subfield $1 or useable $0 present, meaningful IRI is minted to allow for deduplication.

{BASE}{source}/con#{prefLabel}

Example IRI for a skos:Concept without $1 or useable $0:
http://marc2rda.info/testing/lcsh/con#youngwomenenglandfiction

Deduplication

Phase I performs rudimentary entity deduplication through its approach to minting IRIs. Entities with common access points, sources, and entity types are given the same IRI using this method, and are treated as single entities.

While this approach to deduplication minimizes the problem of entity duplication, it runs the risk of conflating similar entities. The Project will revisit deduplication and adopt a more sophisticated approach during Phase II with the goal of minimizing entity conflation.

Reproductions: Mapping and Transformation

Overview

In MARC21 records, a variety of approaches to describing reproductions of earlier published content may appear depending on cataloging rules and practices which have changed over time. Most problematic from the perspective of Official RDA are longstanding practices to include, in one MARC record, information pertaining to the original manifestation such as date of publication, publisher, etc. in records which otherwise describe a reproduction manifestation. The project discussed and ultimately agreed that there was no way to represent our output as RDA without trying to separate out descriptive elements of the original from the record and creating a separate manifestation entity, relating both original and reproduction manifestations to the same expression. For Phase I, we did not attempt to deal with all possibilities for reproductions. However, where a set of rules was used (such as Program for Cooperative Cataloging guidelines), we were able to find markers in MARC21 that signal when an approach has been used, and based on these guidelines, to identify which fields describe the original and which the reproduction. Issues relating to records which follow the practice of describing the original in parts of the record but don't follow PCC guidelines or use the "marker" fields, as well as a comprehensive review of other reproduction cataloging practices, remain for further examination during Phase II.

Documentation Consulted

LC-PCC PS 1.11 Facsimiles and Reproductions, found in RDA Toolkit, Original RDA section (requires subscription to access): https://original.rdatoolkit.org/lcpschp1_lcps1-99044.html

PCC Program for Cooperative Cataloging (Washington, D.C.). Provider-Neutral E-Resource MARC Record Guide: P-N/RDA Version, 2024.
https://www.loc.gov/aba/pcc/scs/documents/PN-RDA-Combined.pdf

Specific Mapping Practices

Minting Entities

- Evaluate whether Condition 1 or Condition 2 (under Conditions for Mapping below) is present in the record. These are markers of a reproduction record that has used a combined original and reproduction descriptive practice. Original and reproduction manifestations must be separated out from such records.

- If either condition is present, mint an IRI for the original Manifestation (OriginalM)
- To relate the OriginalM to the IRI for the Manifestation being described (ReproM):
    - Check the value for Form of Item in 008 (character position 008/23 for Books, Computer Files, Continuing Resources, Music, and Mixed Materials formats; 008/29 for Maps or Visual Materials)
    - If the value is "o" (letter o lowercase, code for online), use property [ReproM] http://rdaregistry.info/elements/m/P30136 //"is electronic reproduction of manifestation of"// [OriginalM]
    - If value is not "o", use property [ReproM] http://rdaregistry.info/Elements/m/P30043 //"is reproduction of manifestation of"// [OriginalM]
    - Relate the OriginalM to the Expression which has been minted for the ReproM, using property http://rdaregistry.info/Elements/m/P30139 //"Has expression manifested"

Conditions for Mapping

Check first for Condition 1. If not present, check for Condition 2.

Condition 1 (provider-neutral online electronic records)

- 588 (Source of description note) if a 588 tag is present in which subfield $a contains the text string "version record."

Condition 2 (microforms, online electronic access, POD and photocopy)

- 533 (Reproduction note) if a 533 tag is present in which subfield $a does not contain the text string "also available as."

Transformation Guidelines

In mapping spreadsheets, under Conditions, the phrase "Check for Reproduction Conditions 1 and 2" are added for all mapping rows which need to be checked (exception: tag 300). A short phrase is inserted into Transformation Notes beginning "If a Reproduction condition is present…".

For most data in 008, 250, 260, 264, and 300, the instruction will be "...apply mapping and transformation to OriginalM, not ReproM."

For 008, Form of Item (008/23 or 29), special instructions conditional on the value present:

- Condition column: "Check for Reproduction Conditions 1 and 2"
- Transformation note: ""If a Reproduction condition is present, AND if value != a, b, c, o, or r, apply mapping and transformation to OriginalM, not ReproM"
- Only a, b, c, o, or r are specified (valid) in PCC instructions for this byte for microforms, electronic reproductions or provider neutral records. These values describe the reproduction. Other values might appear if the record is "cloned" and the byte is not updated from what was in the original manifestation record

For 245, the instruction will be "...apply mapping/transformation to both OriginalM and ReproM". This is because both manifestations need to have a title, which is not supplied in tag 533 and is thus assumed to be the same for original and reproduction.

For tag 300, the Condition column says "Check for Reproduction Condition 2 only", and the Transformation note includes "If a reproduction condition is present, apply mapping/transformation to OriginalM, not ReproM". This is because Provider Neutral instructions say to describe the reproduction in 300, but Facsimiles and Reproductions instructions say to describe the original with regard to extent, etc. (retaining existing field from a cloned PCC record for the original if available).

Upon examination, there seemed to be no value in trying to apply conditional mapping to 8XX fields. From experience with MARC21 records, these may reflect both reproduction series and/or series within which the original was published. We are mapping them to the reproduction manifestation.

Issues Observed with Phase I Transformations Targeted for Improvement in Phase II

Records which don't follow PCC guidelines for reproductions, particularly older, pre-RDA records, need further analysis. This includes records which are describing the original in parts of the record and a reproduction in others, e.g. information about a reproduction residing in a 500 note. Handling of fixed field values may be unpredictable in such records. Large vendor or publisher record sets may present opportunities for pre-transformation remediation.

Information about an original manifestation derived from a MARC record for a reproduction may not be sufficient to construct an access point for the original manifestation or a "matching" type of IRI for that manifestation, that would align with those data points generated from a record for the original. This could prevent a deduplicated description where both original and reproduction converted MARC21 records are in a collection or set of descriptions.

Identifiers

Identifiers for RDA entities taking the form of alphanumeric codes, which are not IRIs (e.g., ISBNs, ISSNs), occur throughout the MARC21 Bibliographic standard. In general, the MARC2RDA Project approaches such identifiers by minting IRIs for Nomen entities to be used as the object for properties such as "has identifier for manifestation", so that context can be given using nomen attributes. Attributes such as "scheme of nomen", "note on nomen", "status of identification" (for invalid identifiers), and others are useful.

Nomen entities are consistently minted for nomen strings for identifiers found in MARC21 data. Aligned with the project's approach to minting other Nomens, if a source is not present identifiers are not treated as nomens but instead are given as string values. Schemes of Nomen are provided, and "status of identification" is used for invalid identifiers (generally found in subfield $z of identifier fields). Paired fields for alternate scripts for identifier nomens are generally not mapped or converted, as they are rare. Control numbers for MARC21 Bibliographic descriptions maintained by an agency are treated as identifiers for the main manifestation described by the record, in the absence of any fields indicating that the number actually applies to another entity. Identifier subfields may be mapped as Nomen attributes, aligned with the Project's approach to 6XX fields.

Subfields of Interest

Subfields $0 and $1

Leveraging the data in legacy MARC21 Bibliographic records as much as possible requires that mapping and transformation processes correctly identify RDA and non-RDA entities hidden in unstructured text. Subfields $0 and $1 have been added to MARC21 to enable the identification of entities referenced in fields. These subfields were not introduced simultaneously and have been used inconsistently to reference either descriptions of entities or the entities themselves (as real-world objects), using either identifiers or full IRIs. Moreover, depending on each library's cataloging policy and choices of preferred value vocabularies, subfields $0 and $1 may point to entities under different models, thereby exhibiting semantics which differ from those of RDA.

For the MARC2RDA Project, the use of identifiers and IRIs found in subfields $0 and $1 depends on the semantics of the vocabularies they belong to. Thus, the challenges have been:

1. To verify whether the IRI from $0 or $1 or the vocabulary from subfield $2 describes entities compliant with LRM/RDA semantics
2. To determine the treatment of entities which are compliant with LRM/RDA semantics, and
3. To determine the treatment of entities which are not compliant with LRM/RDA semantics.

The MARC2RDA Project team has produced an analysis of different vocabularies referred to in subfields $0, $1 and $2, determining their compliance with LRM/RDA semantics. The list of vocabularies can be found in the 'Approved URIs' list. In cases where values in a vocabulary match LRM/RDA semantics, the exact RDA entity (class) is identified, and the IRIs leading to those entities are considered "approved".

Use of an approved vocabulary triggers the selection of object properties during transformation. As a result,

- The IRI that is found in subfield $1 or that is constructed from an identifier in subfield $0 (in the case of FAST for certain entities) is considered the entity IRI
- The entity IRI is used as the object of produced RDA/RDF triples
- A Nomen entity is minted with a nomen string (containing string values found in other subfields) and a scheme of Nomen from information in subfield $2
- The Nomen entity becomes an authorized access point (AAP) for the entity identified by the IRI in subfields $0 or $1

Use of a non-approved vocabulary triggers the minting of a new entity and the use of object properties during transformation. As a result,

- The entity IRI is minted (according to patterns above in "Minting IRIs") and is used as the object of the produced RDA/RDF triple

- The identifier and/or the IRI found in subfields $0 or $1 are stringified as identifiers for the entity
- A Nomen entity is minted with the string values found in other subfields. The Nomen IRI is an opaque one, meaning that it is more or less randomly generated. The scheme of Nomen is declared using information in subfield $2
- The Nomen entity becomes an access point (AP) for the entity identified by the minted IRI

Non-existence of a vocabulary triggers the selection of datatype properties during transformation. As a result, the value of the datatype property is a plain literal.

Special Cases

More than one subfield $1 in a single MARC21 field

When more than one subfield $1 is present in a single MARC21 field where an IRI from an approved vocabulary can be identified, only the first listed 'approved' subfield $1 value is considered during transformation.

Values in 3XX fields

IRIs existing in either subfield $0 or $1 are considered external. Thus, object properties are used, and they take the external IRI as the object of the triple.

skos:Concept IRIs

Subjects and concepts are out of scope for RDA. As a result, the MARC2RDA Project team did not go forward with analysis regarding subject vocabularies for IRIs found in subfields $0 or $1. If an IRI is found in subfield $0 or $1, it is used as the object of an RDA property. Non-IRI FAST identifiers, commonly found in subfield $0, are used as the basis for reconstructed IRIs to be used as values of RDA properties. Other non-IRI identifiers in subfield $0 are not used for subjects or concepts.

Subfield $2

Subfield $2 contains a source code for the source of a value found within the same MARC21 field.
If an IRI is minted for an RDA entity from a MARC21 Headings field, then an IRI for a Nomen is minted for the access point of that entity, and the value of subfield $2 is used to provide the scheme of nomen for that Nomen entity.

If an IRI is minted for a concept (skos:Concept) from a MARC21 classification or subject heading field, then the value of subfield $2 is used to provide the scheme for that Concept.

If an IRI is minted for a concept (skos:Concept) from a MARC 21 field other than the above, then the value of $2 is also used to provide the scheme for that Concept.

Subfield $3

In MARC21, subfield $3 is generally used to indicate a portion of the described resource to which data in a given MARC21 field pertains. Contents of subfield $3 are free text values and do not come from a controlled vocabulary. As such, the MARC2RDA Project treats them as notes. If a subfield $3 is present in a field which maps as an object type (e.g., Content type), then, in addition to a triple for the object, a triple is also generated for subfield $3 as a note.

For example,

336__ $3 Brochure $a text $2 rdacontent

Produces the following triples:

[Expression] http://rdaregistry.info/Elements/e/object/P20001 http://rdaregistry.info/termList/RDAContentType/1020

[Manifestation] http://rdaregistry.info/Elements/m/P30137 "Brochure has Content type: http://rdaregistry.info/termList/RDAContentType/1020"

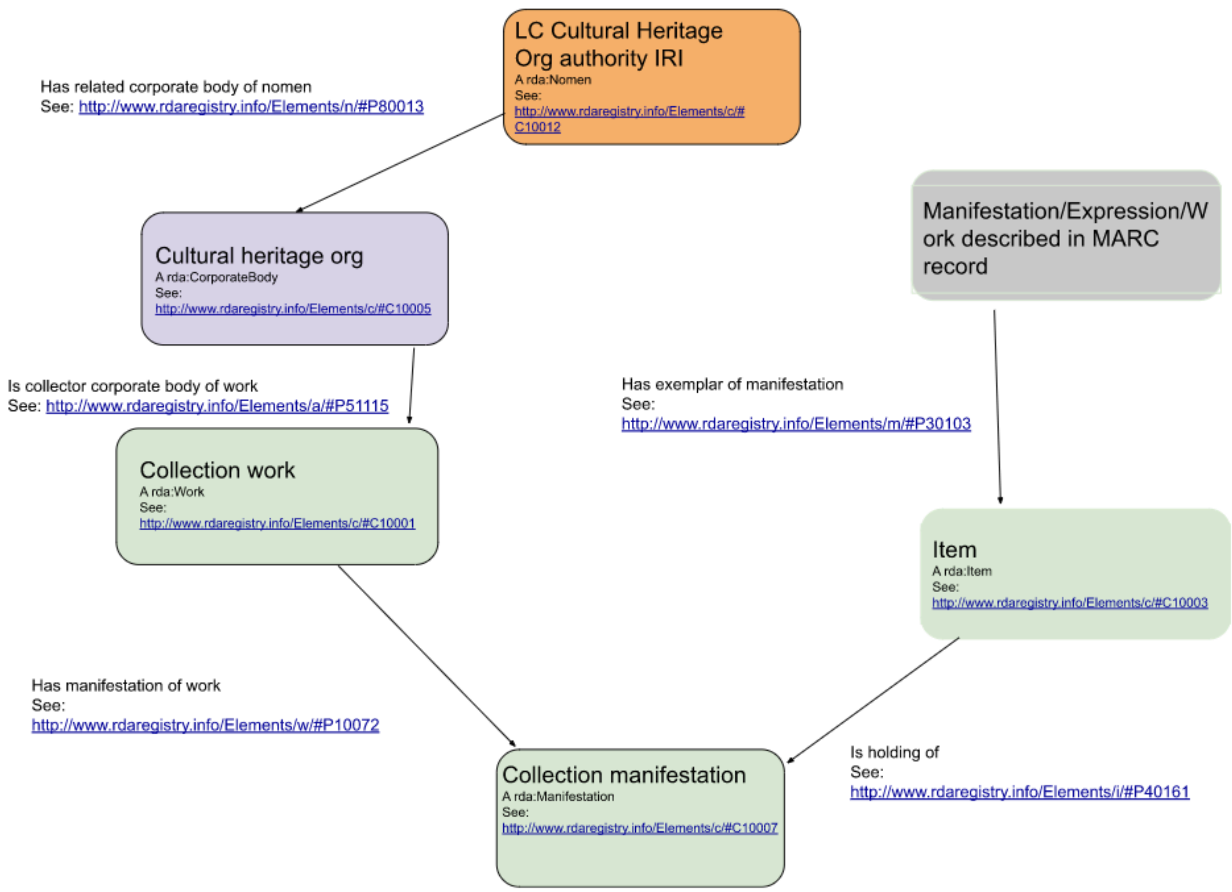
Subfield $5

In MARC21, a subfield $5 generally indicates that the information in a given MARC21 field applies only to a copy held by a specific institution. In RDA, a collection's holdings are conceptualized as Collection Works. Items alluded to in subfield $5 may be holdings of a particular Collection Manifestation associated with the institution whose code is listed as the subfield $5 value.

To facilitate treating information connected to a particular organization's collection in MARC21 Bibliographic data, the MARC2RDA team performed preliminary processing on institution codes for cultural heritage institutions listed in the Code List for Cultural Heritage Organizations vocabulary at id.loc.gov. For each nomen in the vocabulary, a corresponding RDA Corporate Body entity was minted. For each Corporate Body, one Collection Work was minted using boilerplate for appellations based on institution labels in the Library of Congress code list. One Collection Manifestation was minted for each Collection Work. This list of entities was placed in an XML file to be used as a lookup by the transformation code. It is stored in the Project GitHub Repository here.

When the transformation encounters a subfield $5, an Item entity is minted to correspond to the triple that MARC21 field produces. This Item entity is then related to the appropriate Collection Manifestation based on the value of subfield $5. One Item is minted for each occurrence of subfield $5, as there is no way to ascertain which statements from a single institution on a single record refer to the same item or multiple items held by the same institution.

Below is an illustration of our modeling of subfield $5 values in the transformation:



Subfield $6

Subfield $6 is used to retrieve the alternate graphic representation (usually a version in a different script) of a field available in an existing 880 field, and 880 fields are mapped according to the associated field identified in $6. Phase I prioritized fields where $6 and 880 are more commonly present, and associated 880 fields of certain fields, such as classification fields, were not addressed during this phase.

For detailed information regarding the structure of transformed MARC21 Bibliographic data for subfield $6 and corresponding 880 fields, please refer to the project's Decisions Index.

Appendix A: Glossary

Access point: A Nomen that is an appellation of a non-Nomen RDA Entity in natural language that is taken from a vocabulary encoding scheme or is constructed using a string encoding scheme.
Agent: "An entity who is capable of deliberate actions, of being granted rights, and of being held accountable for its actions"--RDA Registry.
Aggregate: "An aggregate is a manifestation that embodies multiple expressions. The distinct expressions that are aggregated may realize one or more works. In addition to the distinct expressions that are aggregated, an aggregate also embodies one and only one aggregating expression that realizes one and

only one aggregating work that is the plan for selecting and arranging the expressions that are embodied in an aggregate. An aggregate may be issued in one or more units. There are three kinds of aggregate: collection aggregate, augmentation aggregate, parallel aggregate. An aggregate may manifest characteristics of more than one kind of aggregate." –RDA Toolkit, September 27, 2025

Augmentation aggregate: "An augmentation aggregate is a manifestation that embodies two or more expressions of two or more works, where one work is supplemented by one or more other works."--RDA Toolkit, September 27, 2025

Authorized access point: "An access point that is selected for preference in a specific application or context."--RDA Toolkit, September 27, 2025.

Coded field (MARC21): MARC21 fields which expect alphanumeric codes, rather than headings constructed using string encoding schemes or unstructured text, as values.

Collection aggregate: "A collection aggregate is a manifestation that embodies two or more expressions of two or more independent works."--RDA Toolkit, September 27, 2025

Concept: In the context of this project, any concept which does not fit into the RDA Entity data model.

Conceptual model: A high-level representation of the semantic modeling serving as the basis for a dataset. RDA uses the IFLA LRM as its conceptual model.

Deduplication: Systematic elimination of excessive copies of data.

Descriptive field (MARC21): A field in MARC21 which does not expect a code or a heading as its value. Rather, descriptive fields contain uncontrolled string values.

Diachronic work: "A diachronic work is a work that is planned to be embodied over time, rather than in a single act of publication. When the plan is carried out, the content of the work changes over time by being realized by one or more discrete expressions that are embodied by one or more manifestations. The essence of a diachronic work is a plan for the change of content."--RDA Toolkit, September 27, 2025

Entity: In linked data, an entity is a real-world object, concept, or thing that is identified by a URI. In RDA, entities are classed into several subclasses of RDA Entity including Works, Expressions, Manifestations, Items, Agents, Timespans, Places, Nomens, and their subclasses.

Expression: "An intellectual or artistic realization of a work in the form of alpha-numeric, musical or choreographic notation, sound, image, object, movement, etc., or any combination of such forms."--RDA Toolkit, September 27, 2025

Field (MARC21): A main component of a MARC21 record, including Leader, Directory, and variable fields.

Group 1 resources: In the MARC2RDA Project, Group 1 resources include manifestations of single expressions and some simple augmentation aggregate manifestations.

Group 2 resources: In the MARC2RDA Project, Group 2 resources include complex augmentation aggregate manifestations, parallel aggregate manifestations, collection aggregate manifestations, and diachronic works.

Heading field (MARC21): A field in MARC21 which is used to construct an access point for an entity.

IRI: Internationalized Resource Identifier. A form of URI which accepts Unicode characters. Used to identify entities in linked data.

Item: "A single exemplar or instance of a manifestation."--RDA Toolkit, September 27, 2025

Literal: In RDF, a literal is a text string.

Manifestation: "A physical embodiment of an expression of a work."--RDA Toolkit, September 27, 2025.

Nomen: "A label for any RDA entity except a nomen. A nomen includes a name, title, access point, or identifier."--RDA Toolkit, September 27, 2025
Ontology: An explicitly-defined, formal data model including RDF classes, properties, and their characteristics within a specific domain. Ontologies are machine-readable rules for implementing conceptual models.
Open source: This project defines open source as a tool or resource which is freely available to anyone, and can be reused for any purpose without permission.
Parallel aggregate: "A parallel aggregate is a manifestation that embodies two or more expressions of a single work."--RDA Registry, September 27, 2025
Place: "A given extent of space."--RDA Registry, September 27, 2025
RDF: Resource Description Framework. A standard of the W3C (World Wide Web Consortium) which provides a framework for representing information on the web in the form of triple statements
RDF class: A category of resources identified as an rdfs:Class. The RDA Registry lists an RDF Class to correspond to each RDA Entity type.
Related entity: For the purposes of the MARC2RDA Project, related entities are those entities which are not the main WEMI entities described for a resource (Resource entities), but are somehow related to those entities.
Reproduction: A manifestation which is a reproduction of another manifestation.
Resource entity: For the purposes of the MARC2RDA Project, resource entities are the main WEMI entities described for a resource. Entities which are not resource entities, but are related to resource entities in some way, are referred to as "Related entities".
String: A set of characters recorded as a literal value.
String encoding scheme: A specific set of rules for constructing string values.
Subfield (MARC21): A component of a MARC21 field which indicates a more specific value for that field. Subfields are identified by letters or numbers, and each letter or number may carry a different meaning depending on the MARC21 field it is used in.
Timespan: In RDA, a Timespan is an entity representing a finite period of time. Open, uncertain, or otherwise "broken" dates are not considered Timespans.
Triple statement: A statement in RDF format comprising a subject, predicate, and object.
Triplestore: "A triplestore or RDF store is a purpose-built database for the storage and retrieval of triples through semantic queries."--Wikipedia, September 29, 2025
WEMI: Short-hand to refer to the main Works, Expressions, Manifestations, and Items associated with a resource in RDA. This project refers to these entities as "Resource entities".
Work: In RDA, a Work is an entity representing a distinct intellectual or artistic creation (as in, the intellectual or artistic content). A Work is distinct from other RDA Entities, and is in conceptual alignment with the LRM Work.
XSLT: Extensible Stylesheet Language Transformations. A programming language designed for processing XML documents.